

‘Colorblind Game’ Can Enhance Awareness of Color Blindness

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Abstract

This study investigates whether awareness of color blindness can be enhanced through a *virtual colorblind gaming experience*. We conducted two user studies—one with undergraduates and one with working adults—using ‘colorblind’ color schemes in a digital game to explore whether such an experience enhances their assessment of their own knowledge about color blindness and awareness of its associated disadvantages in society, workplaces, and private life. The results with undergraduates showed increases in general, but not much in workplace disadvantages. In contrast, the results with working adults showed a significant improvement in the assessment of knowledge but not in other aspects. Thus, a virtual colorblind gaming experience can enhance awareness of color blindness, yet the interpretation of the experience may vary significantly depending on the players’ backgrounds.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

Serious games, color blindness, empathy, awareness, Puyo Puyo

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1 Introduction

Individuals with color blindness, or CVD (color vision deficiency), face various disadvantages. They may encounter social disadvantages, such as legal restrictions in educational opportunities and career choices. Even when such restrictions are not explicitly imposed, they may still face disadvantages in the workplace, such as difficulty understanding color codes/signals or interpreting documents and images/photographs. There are also various disadvantages in private life for them, such as difficulty with tasks like choosing clothes, determining the ripeness/spoilage of fruits and vegetables. Yet, while colorblindness is a prevalent form of visual impairment worldwide, such challenges they may face are often overlooked by those with normal vision.

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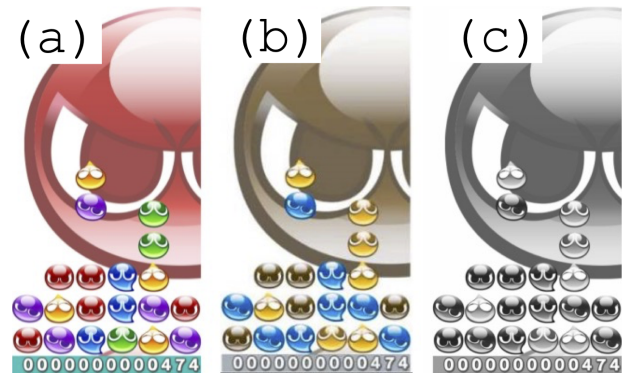


Figure 1: Three versions of Puyo Puyo²: (a) Original, (b) protanopia, and (c) achromatopsia. ©Sega Corporation.

In this paper, we describe our investigation into the research question: “Can a digital game that provides a *virtual colorblind experience* enhance awareness among normal-sighted individuals of the disadvantages faced by people with CVD?” To address this, we created ‘colorblind’ variations of the renowned color-matching puzzle game *Puyo Puyo* and conducted a user study with two populations, undergraduate students and working adults, to evaluate its effectiveness. We particularly contextualize this research question in serious games [Susi et al. 2007].

2 User Study

As mentioned earlier, we developed ‘colorblind’ variations of the popular game, *Puyo Puyo*¹, which simulate the visual perspective of individuals with CVD, as illustrated in Figure 1. Figures 1 (b), and (c) are the screenshots of ‘colorblind version’ of protanopia and achromatopsia. Individuals with protanopia cannot perceive red, and those with achromatopsia cannot perceive any color at all [Daniel 2020]. As *Puyo Puyo* is a colored tile matching puzzle game, such changes in color scheme that emulates color blindness would give a significant damage to its gaming experience. In the experiment, participants were divided into two groups, the *Gameplay* group and the *Document-Only* group. Participants in both groups are first asked to answer a simple questionnaire as described in Table 1, and then to read a brief document explaining CVD and its disadvantages in everyday life. The *Document-Only* group then answered the identical questionnaire again and completed their participation. On the other hand, The *Gameplay* group was asked to play three versions of *Puyo Puyo* (the original and two colorblind versions) illustrated in Figure 1, each for 10 minutes, and then answered the

¹These variations are derived from Sega’s educational materials (<https://edu.monaca.io/puyo>). We sincerely thank Sega Corporation for the permission to use in our study.

Table 1: Four questions asked to the user study participants.

No.	Question
Q ₁	How much do you know about CVD?
Q ₂	Do you think CVD leads to societal disadvantages?
Q ₃	Do you think CVD leads to disadvantages at work?
Q ₄	Do you think CVD leads to disadvantages in private life?

same questionnaire again before completing their participation. All the questions in the questionnaire were responded to using the 10 cm Visual Analogue Scale, where each end represents ‘no, not at all’ and ‘yes, very strongly,’ respectively. Participants marked the point on the scale that best reflected their response to each question. The questionnaire was designed so that participants could compare their answers between the first and second questionnaire, but they were not allowed to modify their answers from the first questionnaire while completing the second one. Participants were allowed to freely interpret *societal disadvantages*, *disadvantages at work*, and *disadvantages in private life* in a manner they considered appropriate, and also to ask any questions if necessary.

First, we performed the user study (*User Study #1*) with 40 undergraduates with normal vision participated. Both the *Document-Only* and *Game Play* groups consisted of 19 males and 1 female. For the reason described in the next section, we then performed the same user study again but with working adults (*User Study #2*). The participants of *User Study #2* were aged between 23 - 50 years old. The participants were divided into 14 males/5 females (*Document-Only*) and 13 males/9 females (*Game Play*), respectively.

3 Results and Discussion

We applied Wilcoxon Signed-Rank Test to determine if there is any statistically significant differences between the answers at the beginning and end of the experiments. The significance level α was set at 0.05 in this analysis. The effect sizes were assessed using Cliff’s delta. As illustrated in Table 2, all the answers in both groups exhibited statistically significant differences with considerably low p-values. Therefore, it can be inferred that the responses provided at the beginning of the questionnaire were lower than those at the end, with statistical significance; in other words, the *virtual colorblind gaming experience* enhanced the awareness of color blindness.

As shown in the result of *User Study #1* in Table 2 (above), a small difference in the effect size of Q₁ (knowledge about CVD) may be attributable to both groups being provided with the same documents that offer information on CVD and its disadvantages. The large differences in effect sizes for Q₂ (societal disadvantages of CVD) and Q₄ (disadvantages in private life) may suggest that the colorblind gaming experience can enhance awareness of the difficulties that individuals with CVD may face. However, it is quite notable that there is only a negligible difference in the effect size for Q₃ (disadvantages at work), unlike in Q₂ and Q₄². We hypothesized that this may be due to the fact that all participants in *User Study #1* were undergraduate students with no professional work experience.

²While the p-values for Q₂ ~ Q₄ in the *Document-Only* group of *User Study #1* are very small, the confidence intervals for the effect sizes include zero. This may suggest that the sample size may not be large enough to evaluate the statistical significance of the effect sizes and/or that the effects for Q₂ ~ Q₄ are statistically insignificant.

Table 2: The results of User Study (95% confidence interval)

User Study #1: Undergraduate Students				
The <i>Game Play</i> group				
Q.	p-value	conf. interval	effect size	conf. interval
Q ₁	p<0.001	-3.10 ~ -1.30	-0.605 (L)	-0.82 ~ -0.23
Q ₂	p<0.001	-1.98 ~ -0.95	-0.415 (M)	-0.69 ~ -0.06
Q ₃	p<0.001	-1.93 ~ -0.63	-0.288 (S)	-0.59 ~ -0.09
Q ₄	p<0.001	-2.13 ~ -0.95	-0.378 (M)	-0.66 ~ -0.00
The <i>Document-Only</i> group				
Q.	p-value	conf. interval	effect size	conf. interval
Q ₁	p<0.001	-2.55 ~ -0.83	-0.528 (L)	-0.76 ~ -0.18
Q ₂	p<0.001	-1.98 ~ -0.95	-0.180 (S)	-0.57 ~ 0.06
Q ₃	p<0.001	-1.93 ~ -0.63	-0.260 (S)	-0.55 ~ 0.08
Q ₄	p<0.001	-2.13 ~ -0.95	-0.292 (S)	-0.59 ~ 0.08
User Study #2: Working Adults				
The <i>Game Play</i> group				
Q.	p-value	conf. interval	effect size	conf. interval
Q ₁	p<0.001	-4.00 ~ -1.90	-0.671 (L)	-0.871 ~ -0.278
Q ₂	p<0.001	-2.65 ~ -0.85	-0.367 (M)	-0.643 ~ -0.007
Q ₃	p<0.001	-2.90 ~ -1.00	-0.396 (M)	-0.660 ~ -0.046
Q ₄	p<0.001	-4.05 ~ -1.30	-0.599 (L)	-0.804 ~ -0.265
The <i>Document-Only</i> group				
Q.	p-value	conf. interval	effect size	conf. interval
Q ₁	p=0.005	-2.85 ~ -0.25	-0.235 (S)	-0.56 ~ -0.15
Q ₂	p<0.001	-3.30 ~ -1.05	-0.429 (M)	-0.70 ~ -0.06
Q ₃	p<0.001	-3.25 ~ -1.20	-0.479 (L)	-0.74 ~ -0.10
Q ₄	p<0.001	-4.65 ~ -2.10	-0.570 (L)	-0.79 ~ -0.21

The effect size is Cliff’s delta, categorized as Large (L), Medium (M), and Small (S).

Hence, we conducted the same experiment with working adults (*User Study #2*). Interestingly, as in Table 2 (below), the result of *User Study #2* was opposite to *User Study #1*. We observed only small differences in the effect sizes of Q₂ and Q₄ but the large difference in that of Q₁ (knowledge about CVD). It is also notable that the effect size decreased in the *Game Play* group in Q₃ (disadvantages at work). This may be attributable to that working adults may have interpreted their experience differently from undergraduates, as they have more life experience and better prior knowledge.

4 Conclusion

We investigated whether *virtual colorblind gaming experience* can enhance awareness of the disadvantages of CVD. The results implied such an experience can be used to enhance the awareness to CVD, while it is necessary to further investigate how the backgrounds of players can influence the interpretation of their experience. Such an issue regarding the interpretation of gaming experience has a potential significance in the research of serious games.

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